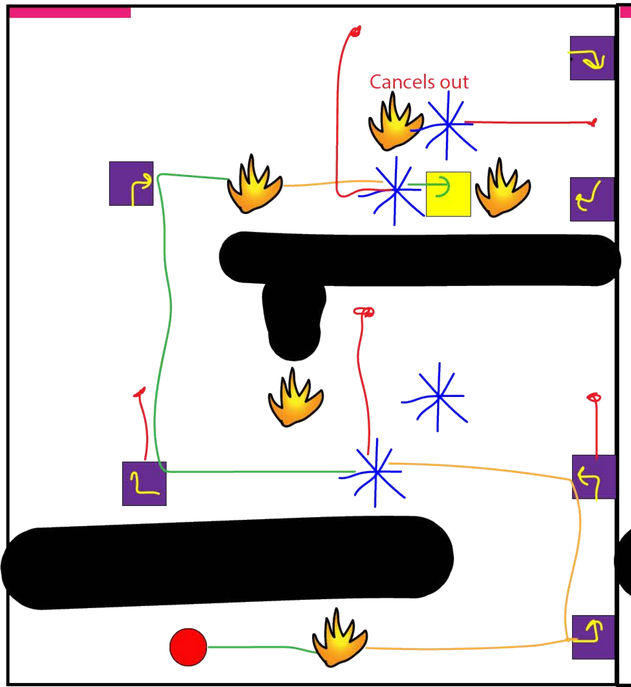
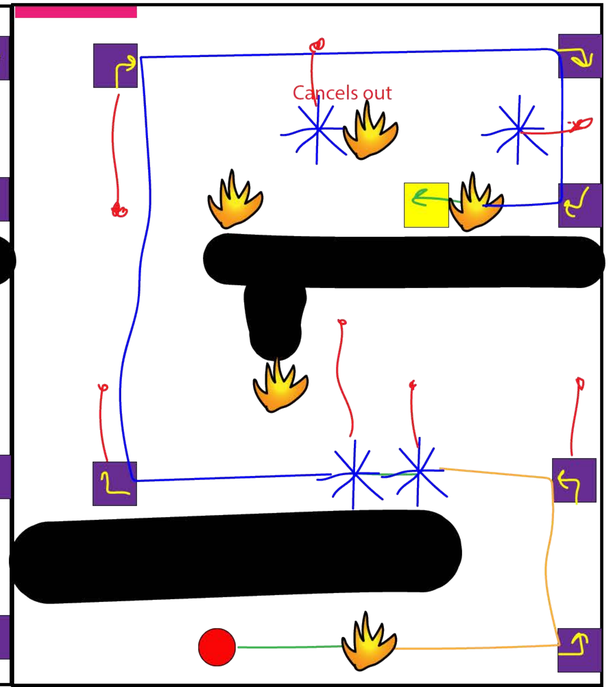
Top-Down Torch Puzzle

Puzzle Description:
The objective is for the player to first open the door and then enter it. To do this, the player must push the ice torches and bounce pads around to ensure that when the arrow collides with the switch, that the arrow isn't fire or ice.

Scenario 1



Scenario 2



Object: Bounce Pad

Properties:
- Is X large
- Is mostly coloured purple

Behaviours:
- Is facing a certain way. This can't be changed.
- One side "takes" an item, another "sends" an item

Relationships:
Dynamic
- When the player interacts with this, the player can push and pull the bounce pad around.
- When arrows pass through the "take" side, the arrow will change direction to face the "send" side. This includes other arrow variants as well.

Object: Fire Arrow

Properties:
- Is X large
- Is coloured brown with a bright orange tip
- Tip has flame animation
- Is on fire

Behaviors:
- Always moving forward at a steady speed

Relationships:
Dynamic
- When colliding with a certain side of the bounce pad the direction is changed.
- when colliding with the ice torch it is changed to a regular arrow.
- When colliding with torch, delete object.

Object: Fire Arrow

Properties:
- Is X large
- Is coloured brown with a bright blue tip
- Tip has frost animation
- Is icy

Behaviors:
- Always moving forward at a steady speed

Relationships:
Dynamic
- When colliding with a certain side of the bounce pad the direction is changed.
- when colliding with the ice torch it is deleted.
- When colliding with torch, change into regular arrow.

Object: Door

Properties:
- Is X large
- Is coloured dark red

Behaviors:
- Static, unmoving

Relationships:
Dynamic
- When the switch is activated, open.
- When player collides with this while in open state, load next area.

Object: Player

Properties:
- Is X large
- Is mostly coloured red

Behaviours:
- Can move in four directions based on player input
- Can swing sword (player input)
- Can shoot arrows (player input)

Relationships:
Dynamic
- Depending on which position the player faces (North, South, West East), the arrow fired will move in that direction.

Spatial
- If the player is within the door's hitbox while it is open, a new area will load.
- The player can push objects like ice torches and bounce pads around as well as pulling them.

Object: Torch

Properties:
- Is X large
- Is mostly coloured orange
- Has a flame animation
- Is on fire

Behaviors:
- Static, never moves

Relationships:
Spatial
- If an arrow passes through this hitbox, the arrow changes to a fire arrow.
- If an ice arrow passes through this hitbox, turn it into a regular arrow.
- If placed right next to Ice torch, becomes unlit torch.
- CANNOT be moved by player.

Object: Ice Torch

Properties:
- Is X large
- Is mostly coloured blue
- Has water flowing animation
- Is icy

Behaviors:
- Static, never moves

Relationships:
Spatial
- If a fire arrow passes through this hitbox, the fire arrow changes to a regular arrow.
- If a regular arrow passes through this hitbox, it becomes an ice arrow.
- If placed next to a torch, become unlit torch.

Object: Switch

Properties:
- Is X large
- Is mostly coloured yellow

Behaviors:
- Static, never moves

Relationships:
Dynamic
- When arrow collides, open door.

Object: Arrow

Properties:
- Is X large
- Is coloured brown with a light grey tip

Behaviours:
- Always moving forwards at a steady speed

Relationships:
Dynamic
- When colliding with a certain side of the bounce pad the direction is changed.
- When colliding with a torch, it changes into a fire arrow.
- When colliding with an ice torch, it changes into an ice arrow.
- When colliding with a switch, open the door.

Object: Unlit Torch

Properties:
- Is X large
- Is colored mostly grey

Behaviors:
- Static, unmoving

Relationships:
Dynamic:
- When colliding with ice arrow, change into ice torch.
- When colliding with fire arrow, change into torch.
- When pushed or pulled by the player, move.

Systemic game with emergent mechanics justification

The reason I designed the arrow to interact with the bounce pad in the way that it does is to make the player use the space they're given differently depending on an unmovable object like the torch. Likewise, the torches have the behavior of being unmovable in order to give the bounce pads a use. This gives the torch and the bounce pad an indirect relationship that isn't explicitly laid out in the rules. The reason that ice torches are movable while regular torches aren't is because then the player could just move everything out of the way of the switch. Having the arrow always interact with the torch at least once means that players will have to use the relationship between ice torches and fire arrows to revert them to normal arrows, so that the regular arrows can interact with the switch. I originally had fountains instead of ice torches that simply reverted fire to regular arrows, and that was it. I changed them to ice torches that turn regular arrows into ice arrows to create a sense of balancing that the player must do to activate the switch. This creates both more potential problems, and more potential solutions.

I only created ten objects in the puzzle, but because of how much interaction the player can have with each object, the amount of branching consequences are vast. For example, solution one, the player moves an ice torch to turn the arrow normal before it inevitably catches fire again, only to be put out once more by an ice torch that couldn't have been there if the torch above it hadn't been already put out by an ice torch. The normal arrow then activates the switch and the door is open. There are many small factors that could completely change this outcome, like if the ice torch hadn't been used to put out the topmost torch, the fire arrow wouldn't have been put out and the switch wouldn't have been activated.